SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1513

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* **(BL2)**

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5 **Total Marks: 25**

Code of Conduct

I Siddhi Srivastava. P (192521454) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Siddhi

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

Name: Siddhi Srivastava
Reg No: 192521454
Course : CSA1513 - Cloud computing

1) Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Sol: Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet.

Four characteristics of cloud computing:

1. On-demand self-service: Users can access computing resources whenever required without contacting the service provider.

   Ex: Creating a virtual machine on Aws within minutes.

2. Broad network access: cloud services can be accessed through the internet using laptops, smartphones, tablets, and other devices.

   Ex: Accessing google drive from a mobile phone or laptop.

3. Resource pooling: Cloud providers share computing resources among multiple customers while keeping their data logically separate.

Ex: A cloud data center serves many companies using shared infrastructure.

4. Rapid elasticity: computing resources can be increased or decreased according to demand.

Ex: An online shopping website.

Difference from traditional on-premise computing:

Traditional computing requires an organization to purchase, install and maintain its own hardware. It reduces initial investment and maintenance responsibilities.

Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

1. Mainframe computing: large, centralized computers served multiple users through terminals. This introduced the concept of sharing computing resources.

2. Personal computers and client-server:

computers became more affordable, nd client-server systems allowed users to ccess resources from central servers.

3. Virtualization: A single physical server could run multiple virtual machines, each with its own operating system.

4. Internet and web services: Faster internet connections and web technologies allowed applications and services to be accessed remotely through browsers and APIs.

5. Utility computing: Computing resources began to be provided like utilities, allowing customers to pay according to their usage.

6. Modern cloud platforms: Providers such as Aws, microsoft azure, and google cloud combined virtualization, large data centers, automation, and internet-based services to offer scalable computing resources.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Sol: Role of server virtualization:

*. Better resource utilization: Multiple virtual machines share one physical server.

*. cost reduction: Fewer physical servers are required.

*. Isolation: Problems in one virtual machine can often be contained without affecting others.

*. Flexibility: Virtual machines can be created, modified or removed quickly.

*. Scalability: Resources can be allocated according to workload requirements.

Virtualization	Cloud computing
It is an enabling technology.	It is a service delivery model.
Creates virtual versions of servers, storage or networks.	Delivers computing resources over a network, usually the internet.
Ex: Running several virtual machines on one physical server.	Ex: Renting a virtual machine through Aws.

Virtualization helps cloud computing use hardware efficiently, while cloud combines the virtualization with networking.

Discuss the role of data centers in delivering cloud services.

Sol: Role of data centers in cloud computing:

1. Computing: Data centers contain servers that process user requests and run cloud applications.
2. Data storage: They store files, databases, backups, and application data.
3. Networking: Network equipment connects servers to the internet and transfers data between users and cloud services.
4. Availability and reliability: Backup systems, redundant equipment, and disaster recovery mechanisms help maintain services during failures.
5. Security: Physical access controls, monitoring, and other security measures help protect infrastructure and data.
7. Power and cooling: Electrical systems and cooling equipment help servers operate continuously and safely.

Data centers form physical foundation of cloud computing by providing the computing power, storage, networking and needed to deliver cloud services.

5) Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Sol:

Service model	Full Form	Description	Real-world example.
IaaS	Infrastructure as a service	Provides virtual machines, storage, and networking.	AWS EC2
PaaS	Platform as a service	Provides a platform for developing, testing and deploying applications.	Google app engine.
SaaS	Software as a service	Provides ready-to-use software over the internet.	Google Docs.
XaaS	Everything as a service.	A broad term for delivering different IT resources and capabilities	Dropbox as storage as a service.

These service models allow individuals and organizations to choose the level of control, flexibility and management responsibility that suits their needs.

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